Boss Challenge — Paper 20 (Class 7)

Acids, Bases & Salts | Weather, Climate & Adaptations | Simple Equations | Symmetry

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A substance that tastes sour and turns blue litmus paper red is called an:

(A) salt
(B) base
(C) acid
(D) neutral substance

2. A statement that two expressions are equal, joined by an equals sign, is called an:

(A) inequality
(B) expression
(C) variable
(D) equation

3. The day-to-day condition of the atmosphere - its temperature, humidity, rainfall and wind - at a place is called its:

(A) climate
(B) humidity
(C) weather
(D) season

4. A line that divides a figure into two identical halves that are mirror images is called a:

(A) diagonal only
(B) perimeter
(C) number line
(D) line of symmetry

5. A substance that feels soapy, tastes bitter and turns red litmus blue is known as a:

(A) acid
(B) base
(C) salt
(D) indicator

6. To solve $x + 5 = 12$, you transpose the 5 and get:

(A) $x = 60$
(B) $x = 5$
(C) $x = 7$
(D) $x = 17$

7. The average weather pattern of a place taken over a long period of about twenty-five years is its:

(A) weather
(B) forecast
(C) temperature
(D) climate

8. The number of lines of symmetry in a square is:

(A) 2
(B) 1
(C) 4
(D) 0

9. When an acid is mixed with exactly the right amount of a base, the reaction is named:

(A) evaporation
(B) oxidation
(C) condensation
(D) neutralisation

10. The solution of the equation $3x = 15$ is:

(A) $x = 5$
(B) $x = 12$
(C) $x = 45$
(D) $x = 18$

11. All the changes we call weather are ultimately driven by energy coming from the:

(A) Moon
(B) stars
(C) Sun
(D) Earth's core

12. A rectangle that is not a square has exactly how many lines of symmetry?

(A) 1
(B) 4
(C) 2
(D) 0

13. The two products always formed when an acid reacts completely with a base are:

(A) a salt and water
(B) two acids
(C) only a gas
(D) a base and oxygen

14. If $x / 4 = 3$, then the value of x is:

(A) $x = 1$
(B) $x = 7$
(C) $x = 12$
(D) $x = 0.75$

15. The amount of water vapour present in the air is measured as the air's:

(A) rainfall
(B) wind speed
(C) temperature
(D) humidity

16. An equilateral triangle (all sides equal) has how many lines of symmetry?

(A) 2

(B) 3

(C) 1

(D) 0

17. Turmeric, a natural yellow dye, turns bright red when it touches a:

(A) acid

(B) base

(C) neutral water

(D) salt

18. Solving $2x + 3 = 11$ gives:

(A) $x = 7$

(B) $x = 4$

(C) $x = 4.5$

(D) $x = 16$

19. The instrument used to measure how much rain has fallen at a place is the:

(A) barometer

(B) anemometer

(C) rain gauge

(D) thermometer

20. The number of lines of symmetry in a circle is:

(A) only 2

(B) infinite (countless)

(C) 4

(D) 0

21. A gardener finds the soil is too acidic for crops. To correct it, the soil is treated with:

(A) plain water

(B) more acid such as vinegar

(C) common salt

(D) a base such as lime (quicklime)

22. 'Five less than a number is 9.' Written as an equation this is $x - 5 = 9$, and the number is:

(A) $x = 14$

(B) $x = 45$

(C) $x = 9$

(D) $x = 4$

23. Animals and plants of a region develop special features that help them survive their climate. These features are called:

- **(A)** habits only
- **(B)** diseases
- **(C)** migrations
- **(D)** adaptations

24. An isosceles triangle (exactly two equal sides) has how many lines of symmetry?

- **(A)** 0
- **(B)** 1
- **(C)** 2
- **(D)** 3

25. An ant's sting injects an acid into the skin. The best soothing rub is a mild:

- **(A)** acid, such as vinegar
- **(B)** base, such as baking soda paste
- **(C)** sugar solution
- **(D)** plain table salt

26. An equation works like a balance: whatever you do to one side, you must:

- **(A)** change only the left side
- **(B)** do exactly the same to the other side
- **(C)** ignore the right side
- **(D)** multiply the left and divide the right

27. Two regions of the Earth with the most extreme climates, studied in your textbook, are the:

- **(A)** hills and plains
- **(B)** rivers and lakes
- **(C)** deserts and oceans
- **(D)** polar regions and the tropical rainforests

28. Among the capital letters A, H, and S, the one that has NO line of symmetry is:

- **(A)** both A and H
- **(B)** H
- **(C)** A
- **(D)** S

29. Lemon juice, orange juice and other citrus fruits owe their sour taste to:

- **(A)** acetic acid
- **(B)** lactic acid
- **(C)** sodium hydroxide
- **(D)** citric acid

30. The equation $5x - 2 = 13$ has the solution:

- **(A)** $x = 2.2$
- **(B)** $x = 3$
- **(C)** $x = 75$
- **(D)** $x = 11$

31. The polar bear's thick layer of fat (blubber) and dense white fur are adaptations that mainly help it to:

- **(A)** keep cool in the heat
- **(B)** change colour with seasons
- **(C)** swim faster than fish
- **(D)** stay warm in the freezing polar climate

32. The capital letter H has how many lines of symmetry?

- **(A)** 2
- **(B)** 0
- **(C)** 1
- **(D)** 4

33. The acid present in vinegar, which we use in cooking and pickles, is:

- **(A)** tartaric acid
- **(B)** acetic acid
- **(C)** hydrochloric acid
- **(D)** citric acid

34. To check that $x = 6$ solves $x + 4 = 10$, you should:

- **(A)** put 6 in place of x and see if both sides are equal
- **(B)** change the 10 to a 6
- **(C)** add 6 and 10 together
- **(D)** rub out the equation and start again

35. A polar bear's white fur also helps it by providing:

- **(A)** warning bright colours
- **(B)** soaking up sunlight
- **(C)** attracting a mate
- **(D)** camouflage against the snow

36. The number of times a figure looks exactly the same as it is turned once around (through 360°) is called its:

- **(A)** perimeter
- **(B)** line of symmetry
- **(C)** order of rotational symmetry
- **(D)** angle of rotation

37. A person suffering from acidity (too much acid in the stomach) is given an antacid, which acts as a:

- **(A)** sour fruit juice
- **(B)** strong acid
- **(C)** neutral salt
- **(D)** mild base

38. Solving the equation $2(x + 3) = 14$ gives:

- **(A)** $x = 4$
- **(B)** $x = 11$
- **(C)** $x = 7$
- **(D)** $x = 8$

39. In the hot, wet tropical rainforest, the bright beak and loud calls of a toucan are adaptations that help it to:

- **(A)** find food and recognise its own kind in the dense forest
- **(B)** survive the freezing cold
- **(C)** fly above the clouds
- **(D)** store water for years

40. A square is turned about its centre through a full circle. Its order of rotational symmetry is:

- **(A)** 1
- **(B)** 4
- **(C)** 8
- **(D)** 2

41. The everyday substance whose chemical name is sodium chloride and which forms when hydrochloric acid neutralises sodium hydroxide is:

- **(A)** lime
- **(B)** baking soda
- **(C)** common table salt
- **(D)** washing soda

42. The value of x in $x/2 + 1 = 5$ is:

- **(A)** $x = 10$
- **(B)** $x = 8$
- **(C)** $x = 4$
- **(D)** $x = 12$

43. Many birds, such as the Siberian crane, fly thousands of kilometres to warmer places when their home turns cold. This seasonal journey is called:

- **(A)** camouflage
- **(B)** hibernation
- **(C)** migration
- **(D)** adaptation of body shape

44. A rectangle (not a square) turned about its centre has rotational symmetry of order:

- **(A)** 1
- **(B)** 2
- **(C)** 0
- **(D)** 4

45. Litmus, the most common laboratory indicator, is a natural dye obtained from:

- **(A)** lichens
- **(B)** lemon skins
- **(C)** turmeric roots
- **(D)** rose petals

46. If 9 = x + 4, then the value of x is:

- **(A)** x = 36
- **(B)** x = 13
- **(C)** x = 9
- **(D)** x = 5

47. A camel can survive the desert because it can go for days without water and has broad feet. The desert climate it is adapted to is:

- **(A)** cool and foggy
- **(B)** warm and rainy
- **(C)** cold and snowy
- **(D)** hot and dry

48. For a square, the smallest angle by which it can be turned so that it looks the same is:

- **(A)** 360°
- **(B)** 90°
- **(C)** 45°
- **(D)** 180°

49. A solution leaves both red and blue litmus papers completely unchanged. The solution must be:

- **(A)** neutral
- **(B)** a weak acid
- **(C)** a strong acid
- **(D)** a strong base

50. 'I think of a number, multiply it by 4, and get 28.' The number is found from 4x = 28, giving:

- **(A)** x = 24
- **(B)** x = 32
- **(C)** x = 7
- **(D)** x = 112

51. Compared with weather, the climate of a place changes:

- **(A)** every single day
- **(B)** only during a storm
- **(C)** very slowly, over many years
- **(D)** from hour to hour

52. A regular hexagon (six equal sides) has how many lines of symmetry?

- **(A)** 6
- **(B)** 4
- **(C)** 3
- **(D)** 12

53. China rose indicator turns a solution dark green. The solution is therefore:

- **(A)** basic
- **(B)** neutral
- **(C)** a salt of a strong acid
- **(D)** acidic

54. The perimeter of a square is 4 times its side. If the perimeter is 36 cm, the side x is found from 4x = 36, so:

- **(A)** x = 144 cm
- **(B)** x = 9 cm
- **(C)** x = 18 cm
- **(D)** x = 32 cm

55. On a clear day, the air temperature at a place is usually highest:

- **(A)** in the early afternoon
- **(B)** at midnight
- **(C)** at sunrise
- **(D)** just before dawn

56. A fusion question. A real snowflake very often shows the same pattern repeating every 60° as it is turned. Its order of rotational symmetry is therefore:

- **(A)** 60
- **(B)** 4
- **(C)** 6
- **(D)** 3

57. Tamarind and grapes get their sourness from:

- **(A)** acetic acid
- **(B)** lactic acid
- **(C)** tartaric acid
- **(D)** citric acid

58. 'The sum of a number and twice the number is 21.' This becomes x + 2x = 21, so the number is:

- **(A)** x = 21
- **(B)** x = 63
- **(C)** x = 7
- **(D)** x = 10.5

59. A maximum-minimum thermometer at a weather station is used to record the:

- **(A)** speed of the wind
- **(B)** humidity of the air
- **(C)** highest and lowest temperatures of the day
- **(D)** amount of rainfall

60. A fusion question. A butterfly with its wings spread looks the same on its left and right halves. The kind of symmetry a butterfly clearly shows is:

- **(A)** no symmetry at all
- **(B)** line (mirror) symmetry, with a vertical line down the middle
- **(C)** rotational symmetry of order 4
- **(D)** infinite lines of symmetry

61. Curd and sour milk taste tangy because they contain:

- **(A)** lactic acid
- **(B)** sulphuric acid
- **(C)** citric acid
- **(D)** tartaric acid

62. Three consecutive whole numbers add up to 18. Taking them as x, x+1, x+2, the smallest number x is:

- **(A)** x = 5
- **(B)** x = 9
- **(C)** x = 6
- **(D)** x = 4

63. The tropical region lies in a broad belt around the Earth's:

- **(A)** equator
- **(B)** North Pole
- **(C)** ocean floor
- **(D)** highest mountains

64. A triangle whose three sides are all of different lengths (a scalene triangle) has lines of symmetry numbering:

- **(A)** 2
- **(B)** 1
- **(C)** 3
- **(D)** 0

65. Toothpastes are usually slightly basic so that they can:

- **(A)** turn the teeth blue
- **(B)** neutralise the acid made by mouth bacteria
- **(C)** dissolve the enamel faster
- **(D)** make the mouth more acidic

66. Solving 6x = 0 gives:

- **(A)** x = 0
- **(B)** x = -6
- **(C)** x = 6
- **(D)** x = 1

67. A weather report says 'humidity is very high today'. This means you should expect:

- **(A)** no clouds to form at all
- **(B)** the air to feel very dry
- **(C)** sweat to dry slowly and the air to feel sticky
- **(D)** sweat to dry instantly

68. A regular pentagon (five equal sides) has lines of symmetry numbering:

- **(A)** 5
- **(B)** 10
- **(C)** 4
- **(D)** 0

69. Phenolphthalein, an indicator, is colourless in acid but turns pink in:

- **(A)** pure water
- **(B)** a base
- **(C)** a salt solution
- **(D)** an acid

70. A fusion question. In a neutralisation, 1 spoon of base neutralises 2 spoons of acid. If 2x spoons of acid are exactly neutralised by 5 spoons of base, then forming 2x = 10 gives:

- **(A)** x = 20
- **(B)** x = 2.5
- **(C)** x = 5
- **(D)** x = 10

71. Places on the sea coast have a more moderate climate than places far inland mainly because:

- **(A)** coasts are always at the equator
- **(B)** the nearby sea warms and cools slowly, easing temperature swings
- **(C)** there is no wind near the sea
- **(D)** the sea blocks all sunlight

72. The mirror image of a shape across a line is also called its:

- **(A)** enlargement
- **(B)** translation
- **(C)** rotation
- **(D)** reflection

73. Factory waste water is often acidic. Before it is released into a river it should be:

- **(A)** neutralised with a base
- **(B)** made more acidic
- **(C)** released without any treatment
- **(D)** heated until it boils

74. A fusion question. A weather station notes that today's maximum temperature is 7 °C higher than the minimum. If the maximum is 31 °C, the minimum t comes from t + 7 = 31, so:

- **(A)** t = 24 °C
- **(B)** t = 7 °C
- **(C)** t = 38 °C
- **(D)** t = 31 °C

75. An elephant's very large ears are an adaptation that mainly help it to:

- **(A)** keep warm in the cold
- **(B)** scare away predators
- **(C)** hear sounds from underground
- **(D)** lose extra body heat and stay cool

76. The capital letter O is usually treated as having:

- **(A)** three lines of symmetry
- **(B)** no lines of symmetry
- **(C)** exactly one line of symmetry
- **(D)** two lines of symmetry (and high rotational symmetry)

77. Milk of magnesia, used to treat acidity, is an example of a:

- **(A)** neutral salt
- **(B)** acid
- **(C)** indicator
- **(D)** base

78. To clear the fraction in (2x)/3 = 4, you first:

- **(A)** add 3 to both sides
- **(B)** subtract 3 from both sides
- **(C)** divide both sides by 3
- **(D)** multiply both sides by 3

79. Weather forecasts that warn of coming rain or storms are especially useful to:

- **(A)** astronomers studying stars
- **(B)** nobody, since weather cannot be predicted
- **(C)** farmers and fishermen planning their work
- **(D)** only television presenters

80. A fusion question. A starfish commonly has five arms spaced evenly around its centre. Turning it, the order of its rotational symmetry is:

- **(A)** 5
- **(B)** 72
- **(C)** 4
- **(D)** 2

81. Soap solution is tested with red litmus paper. You would expect the paper to:

- **(A)** turn colourless
- **(B)** stay red, showing soap is acidic
- **(C)** turn green
- **(D)** turn blue, showing soap is basic

82. A father is 4 times as old as his son. If the son is x years old and the father is 36, then 4x = 36 gives the son's age as:

- **(A)** x = 40 years
- **(B)** x = 144 years
- **(C)** x = 9 years
- **(D)** x = 32 years

83. Penguins in the freezing Antarctic often huddle tightly together in large groups. This behaviour helps them to:

- **(A)** share body heat and keep warm
- **(B)** cool themselves down
- **(C)** fly away from danger
- **(D)** catch more fish on land

84. A general parallelogram (like a leaning rhombus that is not a rectangle) usually has lines of symmetry numbering:

- **(A)** 4 lines of symmetry
- **(B)** 0 lines and no rotational symmetry
- **(C)** 0, but it has rotational symmetry of order 2
- **(D)** 2 lines of symmetry

85. Sting of a wasp is alkaline (basic). The right home remedy is to apply a mild:

- **(A)** sugar paste
- **(B)** acid, such as vinegar
- **(C)** base, such as baking soda
- **(D)** plain water

86. The equation $7x - 3x = 20$ simplifies first to $4x = 20$, so:

- **(A)** $x = 80$
- **(B)** $x = 5$
- **(C)** $x = 2$
- **(D)** $x = 20$

87. Which of these is an element of weather that a weather report regularly includes?

- **(A)** the price of vegetables
- **(B)** temperature, rainfall, humidity and wind
- **(C)** the population of a city
- **(D)** the height of buildings

88. The smallest angle of rotation that brings an equilateral triangle onto itself is:

- **(A)** 90°
- **(B)** 360°
- **(C)** 120°
- **(D)** 60°

89. When sulphur and nitrogen gases from factories dissolve in rain, they make the rain acidic. This polluted rain is called:

- **(A)** a salt shower
- **(B)** hail
- **(C)** dew
- **(D)** acid rain

90. Which of the following is a correct equation (not just an expression)?

- **(A)** $3x - 1 = 8$
- **(B)** $3x - 1$
- **(C)** $5 + 2x$
- **(D)** $7x$

91. The acid-rain that damages the marble of monuments like the Taj Mahal is mainly an example of weather being affected by:

- **(A)** the Moon's gravity
- **(B)** the colour of the soil
- **(C)** the number of birds
- **(D)** air pollution from gases

92. For any regular polygon, the number of its lines of symmetry is always equal to:

- **(A)** the number of its sides
- **(B)** always four
- **(C)** twice the number of sides
- **(D)** zero

93. Pure distilled water is tested with both litmus and phenolphthalein. The result will be:

- **(A)** phenolphthalein turns pink
- **(B)** blue litmus turns red
- **(C)** no colour change at all, because it is neutral
- **(D)** turmeric turns red

94. After solving an equation you find x = 3. Substituting back into 2x + 1 should give:

- **(A)** 7, confirming the solution
- **(B)** 5
- **(C)** 9
- **(D)** 3

95. Red-eyed frogs, monkeys and big-billed birds living together in great variety are a sign of which climate?

- **(A)** the hot, dry desert
- **(B)** the hot, wet tropical rainforest
- **(C)** a snowy mountain top
- **(D)** the cold polar region

96. A figure is said to have rotational symmetry only if its order of rotational symmetry is:

- **(A)** a fraction
- **(B)** exactly 1
- **(C)** more than 1
- **(D)** equal to 0

97. In a neutralisation reaction the heat that is usually given out tells us the reaction is:

- **(A)** a purely physical mixing
- **(B)** endothermic (it absorbs heat)
- **(C)** a change of state only
- **(D)** exothermic (it releases heat)

98. Five identical pens cost 60 rupees altogether. Writing 5x = 60, the cost of one pen x is:

- **(A)** x = 65 rupees
- **(B)** x = 12 rupees
- **(C)** x = 55 rupees
- **(D)** x = 300 rupees

99. A place is described as having a 'hot and humid' climate. The two factors being described are its:

- **(A)** soil type and crop yield
- **(B)** temperature and the amount of water vapour in the air
- **(C)** rainfall and wind direction
- **(D)** air pressure and altitude

100. A kite, which has two pairs of equal adjacent sides, has how many lines of symmetry?

(A) 1

(B) 2

(C) 0

(D) 4